NAME: John Doe | ID: J08819 | DATE: 15/03/24

Q3(a)

Ladder = 10m (hypotenuse c)

Base = 6m (side a)

$c^2 = a^2 + b^2$ (Pythagorean thrm)

$10^2 = 6^2 + b^2$

$100 = 36 + b^2$

$100 - 36 = b^2$

$64 = b^2$

$b = \sqrt{64} = 8$

The ladder reaches 8m up the wall.

Q1(b)

$f(x) = 2x - 5$, $g(x) = x^2 + 1$, find $g(f(3))$

$f(3) = 2(3) - 5 = 6 - 5 = 1$

$g(f(3)) = g(1) = (1)^2 + 1 = 1 + 1 = 2$

Ans = 2

Q3(b)

Base slides 2m further → new base = 6+2 = 8m

$c = 10m,\ a = 8m$

$10^2 = 8^2 + b^2$

$100 = 64 + b^2$

$b^2 = 100 - 64 = 36$

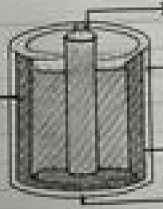
$b = \sqrt{36} = 6$

Iuitial hoight was 8m, now height is 6m.
The tep slides down by 8 - 6 = 2 meters.

Q4

~ Pbotosyatbesis is when plants make food uſing sonligbt and soil.
Photoagathesis: Process by which green plants plants use suolight to sgnthesize nutrients hoos CO_2 and H_2O. Involnes chlornohyll.
$6CO_2 + 6H_2O$ + light energg → $C_6H_{12}O_6$ (Glucose) + $6O_2$

Q8: <u>Dry Cell Diagram</u>



Q7: Newton's Laws

1st - Law of Inertia - An object at rest stags at rest, and an object in motion stags in motion at a constant velocity unless acled upon bg a force. fore. Ev: A passenger jerks forward when a car brakes.

2nd: F = ma (Force = mass x acceleration). Es: More force needed to puch a heavy car than a light car.

3rd: Every action has an equal & opposite reaction. Ex: A rocket launches by pushing gas down.

Q9

(a) Marie Curie, in Physics (for radioactivity in 1903) and later Chemistry (for Radium).

(b) Mars.

Ans 11 (Extra unasked answer)

Q11: Solar System has 8 planets.

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Q11: Solar Syctem has 8 planets.
1. Mercurg, 2. Venus, 3. Earth, 4. Mars...

Q10(a) ROM vs RAM

ROM: Read-Only Memory. Permanent, non-volatile (stores data when power is off). Used for booting (BIOS).

RAM: Random Access Memory. Temporary, volatile (loses data when power is off). Used for currently active programs.

Q10(b) Python function

```
def find_mas(a, b, c):
    if a >= b and a >= c:
        eptorn a
    elif b >= a and b >= c:
        ceturn b
    else:
        return c

pcint(find_mas(12, 45, 7))
```

Q1(a) (Answered last)

$3x^2 - 14x + 8 = 0$

Using quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{(-14) \pm \sqrt{13^2 - 4 \cdot 28}}{2 \cdot 3}$

$x_3 = \frac{14 \pm 10}{6} = \frac{2}{3}$

Roots are $x = 4$ and $x = \frac{2}{3}$.